

SAMPLE STUDENT WORK

The following student work samples are related to the Citizen Math lesson Hair Today, Gone Tomorrow in which students use linear equations to explore the question: How long would it take to donate your hair to make a wig?

In place of these, you could use student work samples from one of your own classrooms. In all cases, you'll want everyone to look at the same work samples, and they should show different valid mathematical approaches for the same problem.

Q1. Each year, thousands of children lose their hair due to cancer or other medical conditions. This can be traumatic. Fortunately, there are organizations that use donated hair to create wigs for sick children.

Imagine the following three people want to donate their hair. For hair to be accepted, it must be at least 10 inches in length. If everyone's hair grows at a rate of 0.5 inches per month*, how many months will it be until each person is eligible to donate?

(*In reality, human hair grows at different rates around the world. According to a 2005 study from the International Journal of Dermatology, people from Africa had hair that grew at a rate of 0.33 inches per month. People from Asia had hair that grew at a rate of 0.53 inches per month. And people from other parts of the world had hair that grew at rates somewhere in between.)

STUDENT A



current length: 0 inches

month	length
0	0 in
1	0.5 in
2	1.0 in
3	1.5 in
4	2.0 in
5	2.5 in
10	5.0 in
20	10.0 in



current length: 8 inches

month	length
0	8.0 in
1	8.5 in
2	9.0 in
3	9.5 in
4	10.0 in



current length: 12.5 inches

month	length
0	12.5 in
-1	12.0 in
-2	11.5 in
-3	11.0 in
-4	10.5 in
-5	10.0 in

Q1. ...Imagine the following three people want to donate their hair. For hair to be accepted, it must be at least 10 inches in length. If everyone's hair grows at a rate of 0.5 inches per month*, how many months will it be until each person is eligible to donate?

STUDENT B

		
$10 - 0 = 10 \text{ inches}$	$10 - 8 = 2 \text{ inches}$	$10 - 12.5 = -2.5 \text{ inches}$
$10 = 0.5t$	$2 = 0.5t$	$-2.5 = 0.5t$
$t = 20 \text{ months}$	$t = 4 \text{ months}$	$t = -5 \text{ months}$

STUDENT C

		
$0 + 0.5m = 10$	$8 + 0.5m = 10$	$12.5 + 0.5m = 10$
$0.5m = 10$	$0.5m = 2$	$0.5m = -2.5$
$\frac{0.5}{0.5} \quad \frac{10}{0.5}$	$\frac{0.5}{0.5} \quad \frac{2}{0.5}$	$\frac{0.5}{0.5} \quad \frac{-2.5}{0.5}$
$m = 20$	$m = 4$	$m = -5$

Q1. ...Imagine the following three people want to donate their hair. For hair to be accepted, it must be at least 10 inches in length. If everyone's hair grows at a rate of 0.5 inches per month*, how many months will it be until each person is eligible to donate?

STUDENT B



current length: 0 inches

$$\text{slope} = 0.5 \text{ in/month}$$

$$y\text{-int} = 0 \text{ inches}$$



current length: 8 inches

$$\text{slope} = 0.5 \text{ in/mo.}$$

$$y\text{-int} = 8 \text{ inches}$$



current length: 12.5 inches

$$\text{slope} = 0.5 \text{ in/mo.}$$

$$y\text{-int} = 12.5 \text{ inches}$$

HAIR LENGTH & HAIRCUTS

